ISRAT JAHAN CHOWDHURY

SOFTWARE ENGINEER

Dhaka, Bangladesh • 01710443216 • isratjahanchowdhury07@gmail.com

github.com/IsratJahanChowdhury • researchgate.net/profile/Israt-Chowdhury-5

PROFILE

CSE undergraduate at American International University–Bangladesh with a 3.82/4.00 CGPA and hands-on experience in web and backend development through academic projects. Skilled in NestJS, TypeScript, JavaScript, PHP, PostgreSQL, MySQL, and Git. Passionate about building practical web applications, learning new technologies, and contributing to collaborative development teams.

EDUCATION

B.Sc. in Computer Science & Engineering — American International University–Bangladesh (AIUB)

CGPA: 3.82/4.00 • Expected Graduation: 2027

TECHNICAL SKILLS

Languages: C++, Java, C#, Python, PHP, JavaScript, HTML, JSON

Backend: NestJS, TypeScript, REST APIs, DTO, class-validator, JWT, Swagger

Databases: SQL, PostgreSQL, MySQL, Oracle, TypeORM

Data & Office: Data Cleaning, Microsoft Office, Microsoft Excel

Tools: Git, GitHub, Postman, VS Code, Figma, Trello

Frontend & Networking: React, Cisco Packet Tracer

OS: Windows

SOFT SKILLS

Communication • Teamwork • Time Management • Leadership

PROJECTS

DineSpace — Multi-Restaurant Management & Ordering Platform

Backend — Reservation Module | Group Project • Tech: NestJS, TypeScript, PostgreSQL, TypeORM, JWT, Swagger, class-validator

- Built the backend reservation workflow in a modular NestJS architecture, connecting reservation logic to PostgreSQL through TypeORM entities and repositories.
- Used DTOs and class-validator for request validation, JWT for authenticated access, and Swagger for API documentation.
- Added search, filtering, sorting, pagination, and logging to support maintainable reservation services.

Hospital Management System

Patient Role | Group Project • Tech: Software Engineering, UML, Figma, Git/GitHub, Trello

- Implemented the Patient-side workflow for a role-based hospital platform covering appointments, admissions, medical records, medicines, tests, complaints, and billing.
- Translated patient requirements into workflows for booking/cancelling appointments, admission requests, medical history/prescriptions, medicine orders, test booking, and bill payment.
- Worked with Waterfall planning, UML/class/activity/DFD models, Figma wireframes, Git/GitHub workflow, and testing plans.

Item Rental Management System

Renter Part | Group Project • Tech: PHP, MySQL, HTML, CSS, JavaScript, AJAX, MVC

- Developed renter-side workflows in an MVC-based rental platform connected to MySQL.
- Implemented flows around item discovery, booking, cancellation, complaints, refund requests, payments, and reviews.
- Worked with session-based user flows and database-driven operations.

Birth Record Management System

Database Design & Documentation | Group Project • Tech: Oracle SQL, ERD, Normalization, Relational Algebra

- Designed and documented a relational database for children, parents, hospitals, doctors, deliveries, and health records.
- Performed 1NF, 2NF, and 3NF normalization before final table creation.
- Created SQL queries using joins, subqueries, aggregate functions, views, and sequences, and wrote relational algebra expressions.

AIUB Smart Service Hub

Requirements & System Design | Group Project • Tech: UML, Requirements Engineering, Trello, Figma

- Contributed to a centralized student-service platform covering tutor booking, Lost & Found, marketplace, clubs/events, emergency support, and an AI assistant.
- Prepared the problem statement, solution description, solution-to-problem analysis, UML diagrams, and social-impact section.

Computer Graphics Game

Stage 5 | Group Project • Tech: C++, OpenGL, GLUT

- Implemented Stage 5 gameplay logic and interactive elements within the team's multi-stage OpenGL game.

Owsudh Chai

Patient Role | Group Project • Tech: Java, Java Swing, OOP

- Developed the Patient role and patient-facing screens and workflows using object-oriented and event-driven GUI concepts.

RESEARCH & PUBLICATIONS

Comparative Evaluation of Deep Learning Architectures for Underwater Semantic Segmentation Using the SUIM Dataset

3rd Author • Published • ICIICE 2026, Dubai, UAE • IEEE. Comparative study involving SegFormer, SETR, Xception, GhostNet, Detectron2, and YOLOv11, using preprocessing, augmentation, transfer learning, fine-tuning, and evaluation with Pixel Accuracy, Mean IoU, Dice Coefficient, and ROC-AUC.

SUBMITTED RESEARCH

A Division-Sensitive Machine Learning Framework for Climate-Based Crop Yield Prediction in Bangladesh

2nd Author | Submitted | 2025

Short Summary: Developed a machine learning framework for Boro rice yield prediction in Bangladesh using long-term agro-climatic data from 2000–2024.

- Evaluated Extra Trees, Random Forest, Gradient Boosting, and XGBoost regression models.
- Applied feature engineering, lag-based yield features, robust scaling, hyperparameter tuning, and stratified 5-fold cross-validation based on regional divisions.
- Performed overall and division-wise evaluation using R^2, RMSE, and MAE.
- Extra Trees Regressor achieved the best overall testing performance with $R^2 = 0.9220$ and RMSE = 0.1612.

Heart Attack Risk Stratification Using Clinically Inspired Derived Labeling and Supervised Learning

Submitted | ICCA 2026 | April 2026

Short Summary: Healthcare AI research exploring heart attack risk prediction using clinically informed labeling and supervised learning techniques.

- Contributed to machine learning-based cardiovascular risk assessment research.
- Investigated Logistic Regression, Random Forest, and XGBoost models.
- Evaluated models using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Cross-Validation.
- Explored clinically informed labeling approaches to improve prediction reliability.

Design and Realization of a Moveable ECG Device Utilizing STM32F1 ARM Microcontroller

1st Author | Submitted | 2025

Short Summary: Designed a portable cardiac monitoring system using STM32F103C8T6 for real-time ECG and heart-rate monitoring.

- Integrated the AD8232 ECG sensor and MAX30102 heart-rate sensor to capture real-time cardiac parameters.
- Implemented real-time visualization of ECG signals and heart-rate data on an OLED display.
- Focused on a cost-effective and scalable cardiac monitoring solution for underserved and inaccessible areas.
- Compared the proposed system with existing cardiac monitoring solutions based on features and implementation cost.

ONGOING RESEARCH

Software Vulnerability Detection Using Graph Neural Networks

Ongoing Thesis | AIUB | February 2026 – Present

Short Summary: Research focused on identifying software vulnerabilities through graph-based machine learning approaches.

- Investigating Graph Neural Networks (GNNs) for automated software vulnerability detection.
- Exploring graph representations of source code for security analysis.
- Preparing research findings for international publication.

VOLUNTEER EXPERIENCE

Official Volunteer — AIMD Symposium 2026

Artificial Intelligence and Molecular Design Symposium 2026

- Provided participant support by assisting attendees with event-related guidance and information.